Abnormal Constituents Of Urine

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MBBS

Learning Objective

Students should be able to

- Interpret normal urine
- Enlist abnormal constituents of urine
- Correlate the abnormal constituents with clinical conditions

Urine

Urine is a metabolic waste product filtered by the kidneys and excreted from the body.

Approximately 1200 ml of blood i.e. 25 % of cardiac out perfuse the kidneys each minute in normal healthy adult.

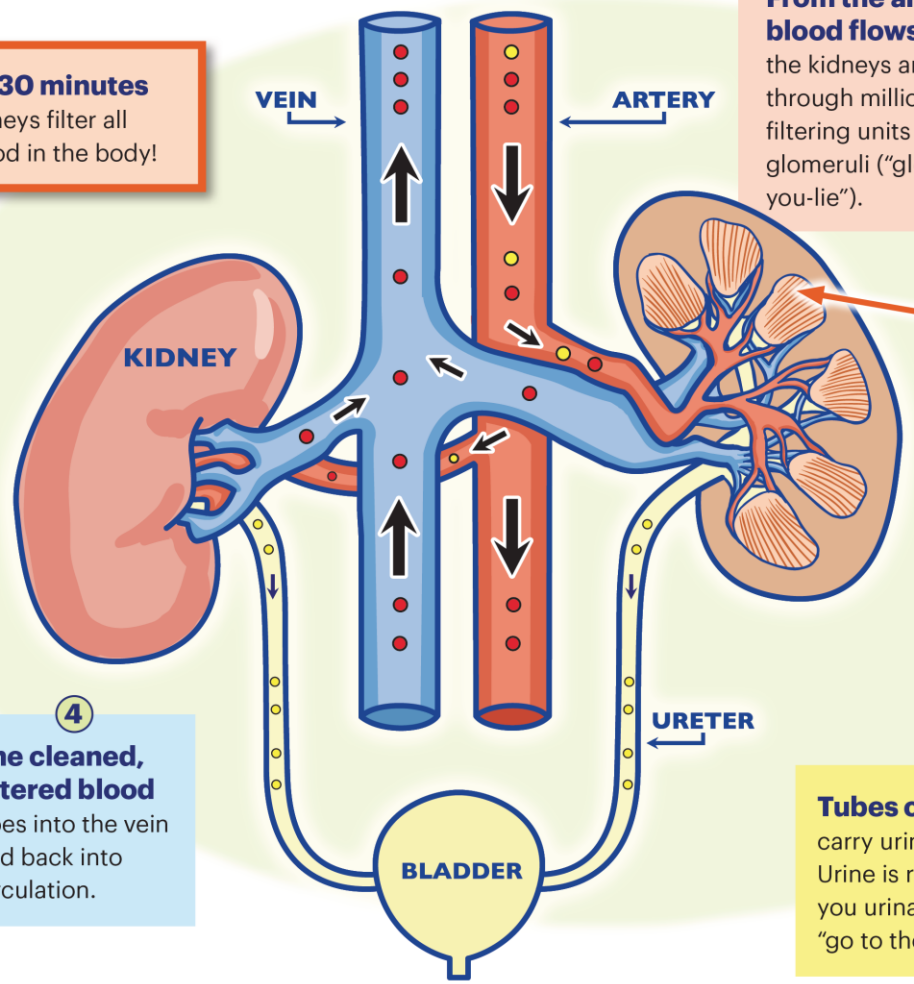
Urine examination helps in the diagnosis of various renal as well as systemic disease.

Urine collection

- Urine is collected in a sterile mouthed container
- Different methods of urine collection are
  - First morning sample(conc. Urine for biochemical analysis ,casts and crystals)
  - Random Sample(chemical screening and microscopic examination)
  - 24 hour urine sample(quantitative estimation of sugar , protein , electrolyte and hormones)
  - Mid stream urine
  - Clean catch urine

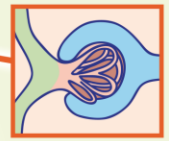
Healthy Kidneys

Every 30 minutes the kidneys filter all the blood in the body!



Blood Cells
Waste

1 **From the artery, blood flows into** the kidneys and passes through millions of tiny filtering units called glomeruli ("glow-mare-you-lie").



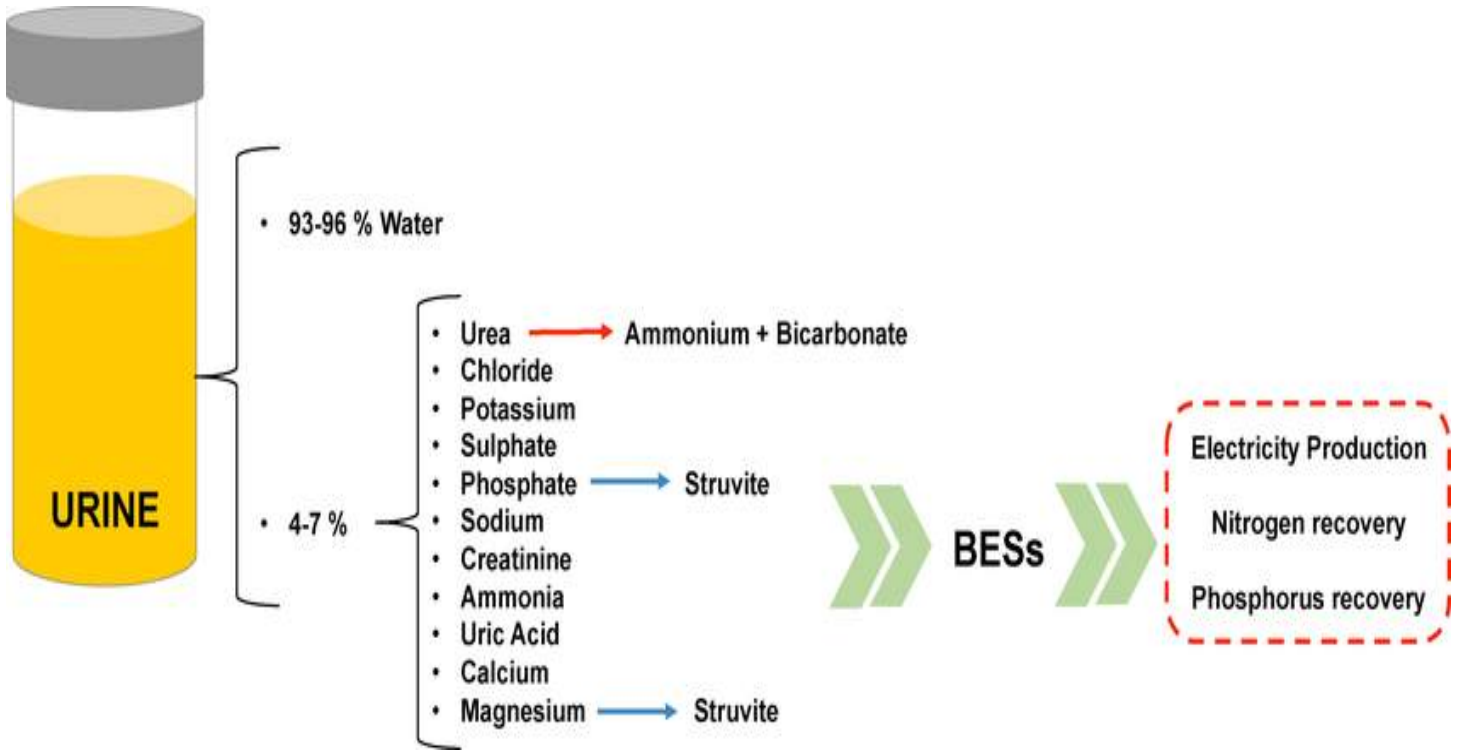
Enlarged glomeruli

2 **Glomeruli can tell** the difference between something you need and something you do not need. They filter out waste and extra fluid from the blood to make urine.

3 **Tubes called ureters** carry urine to the bladder. Urine is removed when you urinate ("pass water," "go to the bathroom").

4 **The cleaned, filtered blood** goes into the vein and back into circulation.

Constituents of urine

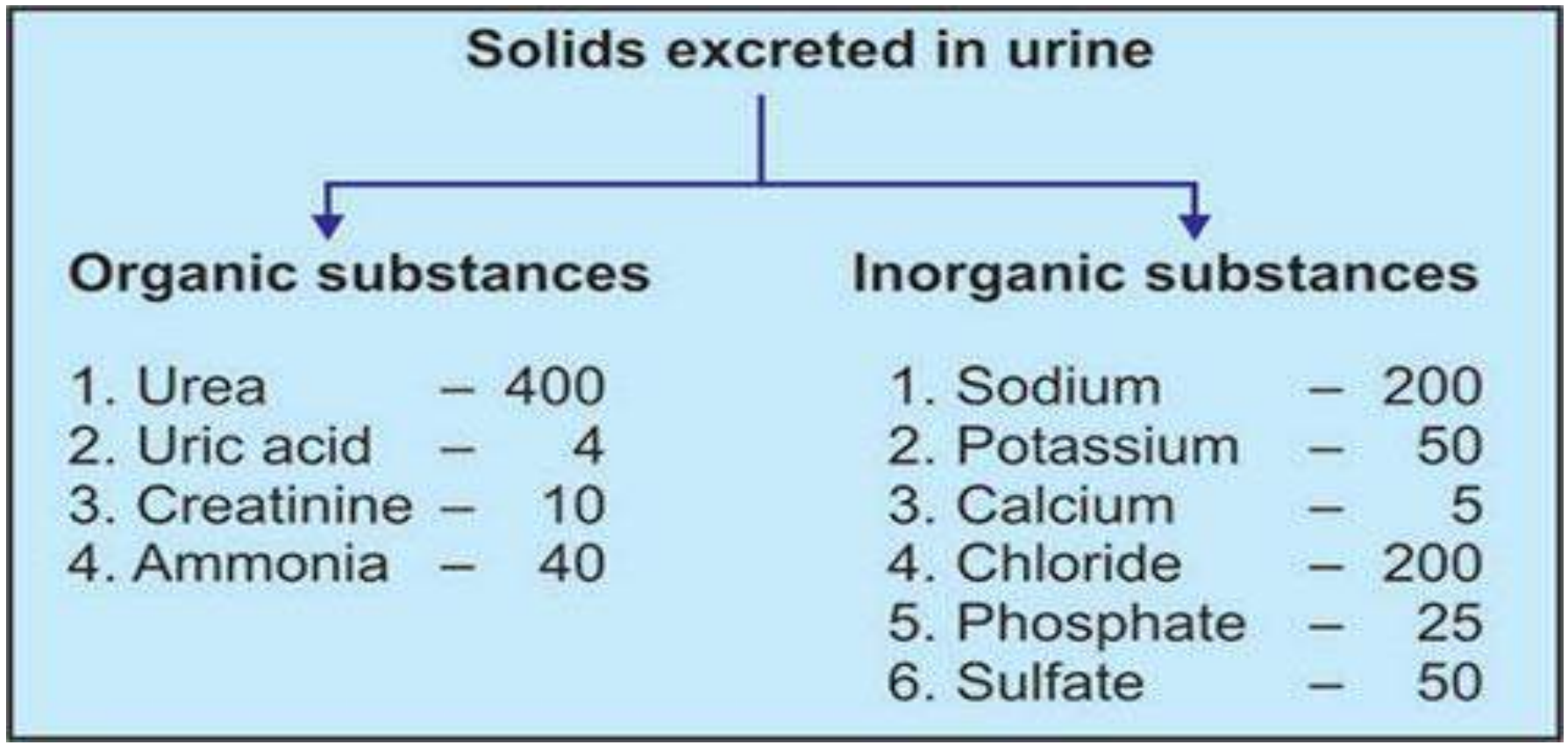


Composition of Normal Urine



Constituents	Amount present
Water	95%
Urea	9.3 to 23.3 gram/liter
Chloride	1.87 to 8.4 gram/liter
Sodium	1.17 to 4.39 gram/liter
Potassium	0.750 to 2.61 gram/liter
Creatinine	0.670 to 2.15 gram/liter
Sulfur	0.163 to 1.80 gram/liter

iClinic


Solids excreted in urine
Organic substances
1. Urea – 400
2. Uric acid – 4
3. Creatinine – 10
4. Ammonia – 40
Inorganic substances
1. Sodium – 200
2. Potassium – 50
3. Calcium – 5
4. Chloride – 200
5. Phosphate – 25
6. Sulfate – 50

Physical properties

- During gross physical examination we observe:

1. Volume
2. Colour / Appearance
3. Smell
4. pH
5. Specific gravity

volume

- Normal urine of volume excreted per day is 1000 – 2000ml / day.
- Night urine output < 400 ml
- Factors which influence the volume excreted
- Intake of fluids, proteins and salt.
- Excessive perspiration and Strenuous exercise decrease the urine output.

- **Normal Volume:** 1-2 litres of urine / day
- **Polyuria:** > 2000 ml of urine / 24 hours

Causes:

- Diabetes Mellitus
- Diabetes insipidus
- Tubular necrosis
- Certain tumors in brain and spinal cord etc.

- **Nocturia:** Excessive urination at night
- **Causes:**

- Enlarged prostate
- Drugs and medications
- Urinary tract infection
- Diabetes
- Obstructive sleep apnea

- **Oliguria:** Excretion of < 500 ml of urine per 24 hours
- **Causes:**

- Glomerulonephritis
- Renal Ischemia
- Dehydration (prolong vomiting, diarrhea)
- Obstruction of urinary tract

Colour

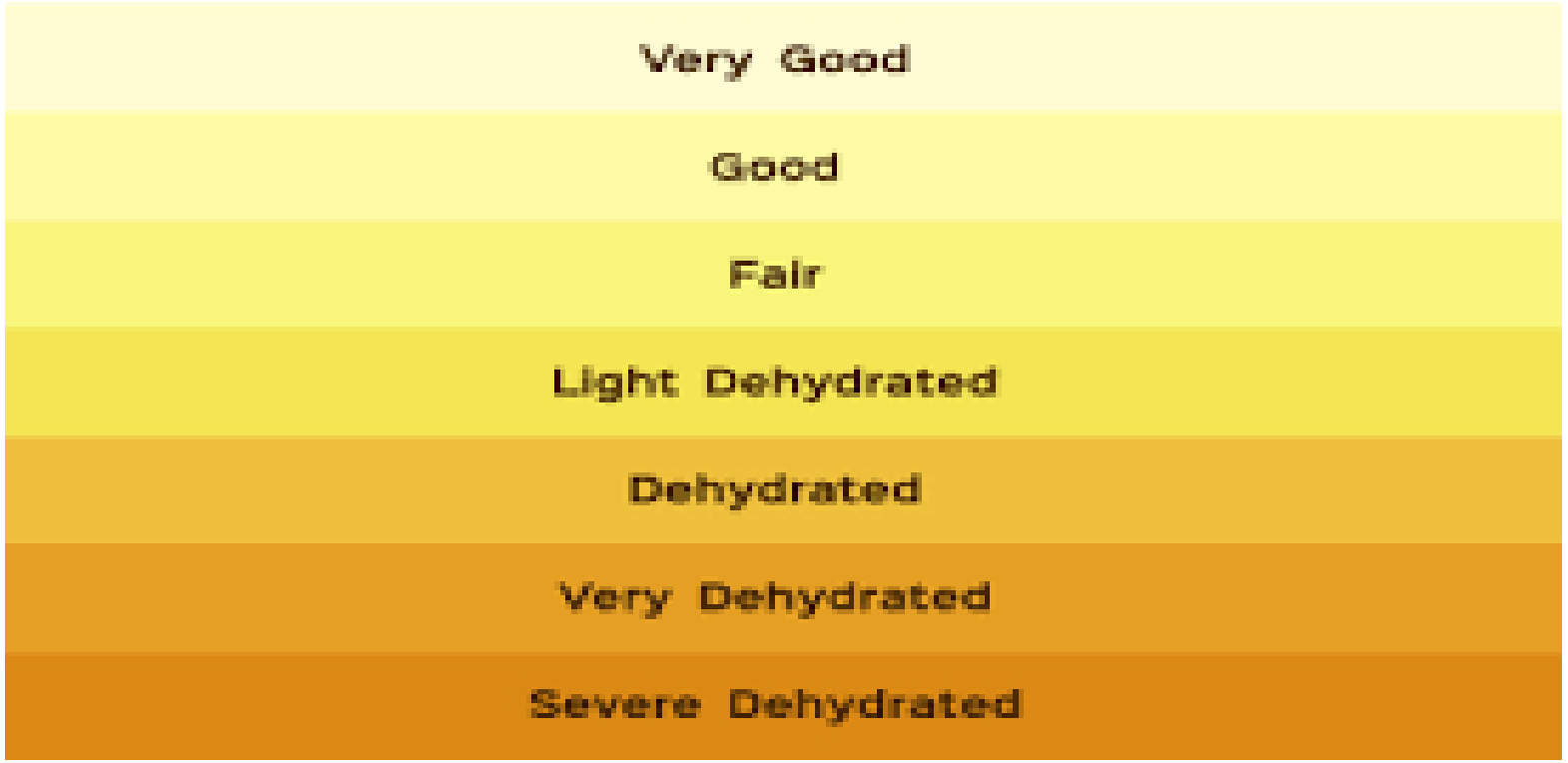
- Normal urine is pale yellow (due to presence of pigment urochrome).
- When the output of urine is low it appears deep yellow.
- Freshly voided urine is clear and transparent.
- Long standing urine may become turbid (due to precipitation of phosphate)

Appearance:

Color of urine

Color	Cause
pale yellow to amber color	Normal colour of urine
Orange color	Due to pigments urochrome Urobilin,porphyrins
Yellow – Brown or Green brown	Bile pigment chiefly bilirubin-biliverdin level
Brownish black	Methemoglobin, Homogentisic acid
Pink to Red	Hemoglobin, Erythrocytes, Myoglobin, Porphyrin, Menstrual contamination.
Cloudy or turbid	Bacteria, pus, mucus, epithelial cells, fats, phosphate and urate

URINE COLOR


Very Good
Good
Fair
Light Dehydrated
Dehydrated
Very Dehydrated
Severe Dehydrated


Color of Urine
Urine Colors
www.labtestsguide.com

Odor

- Normal odor is aromatic in nature

- Long standing urine may have an ammoniacal odor (due to decomposition of urine to ammonia)

pH

- Normal pH of urine ranges from 5.5 – 6.5
- Factors which influence ph are
- High protein diet makes urine more acidic
- Diet rich in vegetables and fruits make urine more alkaline.
- It is measured by urine dipstick, litmus paper or ph paper

Specific gravity

- It is measurement of the kidneys ability to concentrate urine. It is directly related to the amount of solutes in the urine.
- Normally it ranges from 1.010 to 1.025.
- Measured using urinometer, refractometer, dipstick.

chloride

- Normal chloride excreted in 24 hours urine sample is 8 – 15 g/ day (NaCl).

Causes of chloride imbalances

- Hyperchloremia
  - Dehydration
  - Cushing's syndrome
  - Kidney disease
  - Metabolic Acidosis
  - Respiratory Alkalosis
  - Diarrhea
  - over activity of the parathyroid glands
  - Polypharmacy (diuretics, carbonic anhydrase inhibitors, & hormone treatments are known contributors to hyperchloremia)
- Hypochloremia
  - occur with any disorder that causes low blood sodium
  - prolonged vomiting or gastric suction,
  - emphysema or other chronic lung diseases (causing respiratory acidosis)
  - metabolic alkalosis
  - Drugs such as: bicarbonate, corticosteroids, diuretics, and laxatives
  - prolonged diarrhea, sweating or high fevers

Sulphate

- Daily excretion = 1g / day.
- Sulphur is derived from catabolism of sulphur containing amino acids.

Increased level due to

- Homocysteinuria
- Cystenuria
- High protein diet

Decreased level due to

- Renal impairment

Calcium

- Daily excretion – 200 mg / day

Increased level due to

- Hyperparathyroididm
- Renal stones
- Hypervitaminosis D
- Multiple myeloma

Decreased level due to

- Tetany

INORGANIC PHOSPHATE

Daily excretion = 1gm / day

Increased levels due to

- Rickets/ osteomalacia
- Hyperparathyroidism
- Acidosis

Decreased levels due to

- Diarrhoea
- Nephritis

- Pregnancy
- Parathyroid hypofunction

Ammonia

- Daily excretion 0.5 – 0.8/ day.
- Urinary ammonia is derived from other amino acids by kidney.

Increased levels due to

- Ingestion of acid forming foods
- Diabetic ketoacidosis
- Urinary tract infection

Decreased levels due to

- Alkalosis
- nephritis

Normal Organic Constituents of urine

Urea

- Daily excretion 20 – 30 gram / day.
- Urea is formed in the liver.
- End product of protein catabolism.

Increased levels due to :

- Fever
- Diabetes Mellitus

Decreased level due to:

- Liver disease
- metabolic respiratory acidosis nephritis
- nephritis

Uric Acid

- Daily excretion 0.6 – 1 gm / day
- End product of purine catabolism
- Increased levels of uric acid in urine is uricosuria.

Increased levels due to

- Leukemia
- Cancers

Decreased levels due to

- Purine free diet
- Gout

Creatinine

- Daily excretion 2gm / day in males 1gm / day in females
- Urinary creatinine is formed from muscle creatinine.

Increased levels due to

- Muscular disorders
- Myasthenia Gravis
- Thyrotoxicosis
- Stravation
- Uncontrolled diabetes mellitus

Decreased levels due to

- Renal Failure

UROBILINOGEN

- Increased urobilinogen concentration is urine is a sensitive index of liver dysfunction or hemolytic disorder.
- Urobilinogen is present in excessive amount in prehepatic and hepatic jaundice.
- Urobilinogen is absent in urine in post hepatic jaundice.

Dr. Essa Laboratory
& Diagnostic Centre
ISO 9001:2015 (MC) QMS Certified Organization

LOC: North Karachi

CLINICAL INFORMATION/COMMENTS

MR. No. 21061831419

lab No.	2106990986	**Collection:**	01-06-2021 16:40
Name:	MR SUALEH ANSARI		
Gender:	Male	**Age:**	56 Years
Contact:	03002587898	**CNIC/Passport:**	
Ref. By:	02 ON PATIENT REQUEST		
Status:			

Parasitology (Urine D/R)

Test Name	Current Result	Previous Result	Unit	Normal Range
Physical Examination	-			
* Colour	Yellow			Yellow
* Appearance	Hazy			Clear
* Specific gravity	1.005			1.001 - 1.030
* Reaction pH	8.0			5.0 - 8.5
Chemical Examination	-			
* Protein	Negative			Negative
* Glucose	Negative			Negative
* Ketone bodies	Negative			Negative
* Bilrubin	Negative			Negative
* Urobilinogen	Normal			Normal
* Nitrite	Negative			Negative
* Blood	Negative			Negative
* Leucocytes esterase	Negative			Negative
Microscopic Examination	-			
* Pus cells	4 - 6		/hpf	< or= 5
* RBCs	Occassional		/hpf	< or= 5
* Epithelial cells	Occassional		/hpf	< or= 5
* Mucus Threads	+		/hpf	None seen
* Crystals	Ammorphous phosphate + + +		/hpf	None seen

This is a computer generated report and does not require a signature.

Op.
Date: 01/06/2021 09:40:46 PM

Urinalysis with urine C/S

Result Name	Results	Units	Reference Range
Color	Yellow		Yellow
Clarity	Clear		Clear
Bilirubin, urine	Negative		Negative
Specific gravity	1.011		1.003-1.030
Occult blood	Small		Negative
pH	7.5		4.5-8.0
Urobilinogen	1.0	EU/dl	0.2-1.0
Nitrate	Negative		Negative
Leukocytes	Trace		Negative
Glucose	500 (1/2%)	mg/dL	Negative
Ketones	Negative	mg/dL	Negative
Protein	100 (2+)	mg/dL	Negative
WBC	4-10	/hpf	Negative
RBC	4-10	/hpf	Negative
Epithelial cells	None seen	/hpf	None seen
Bacteria	Occasional	/hpf	None seen
Urine culture	No growth after 3 days		

Abnormal constituents of urine

Albumin

- Presence of protein in urine- proteinuria.
- Glomerular walls are impermeable to albumin.
- Kidney disorders show albumin presence in urine. Albuminuria.
- Causes-
- Glomerular nephritis , pyelitis, urethritis, nephrosis etc.
- Heart diseases, fever, anemia, prolonged hypotension.

- Orthostatic albuminuria

  Found in persons standing long term in standing posture.
- Athletic albuminuria

  Found in athletes during vigorous practice.
- Adolescent albuminuria

  found in growing boys

Glucose

- Major constituents of glomerular filtrate
- Reabsobred after glomerular filtration
- In hyperglycemic patients – reabsorption decreases
- Presence of glucose in urine – Glycosuria.
- Symptom of diabetes mellitus.

GLUCOSURIA

SIGNO



Las moléculas de **glucosa** logran filtrarse con facilidad por el glomérulo, por lo que tienen que reabsorberse en el **túbulo proximal**. Si hay demasiada glucosa en sangre, esta no podrá ser reabsorbida en su totalidad, escapando por la orina, causando el signo característico en la **diabetes**.

Para controlar la glucosuria "controle la hiperglucemia"

La causa más común de **glucosuria** es por hiperglucemia, que supera el umbral de reabsorción a nivel del túbulo contorneado proximal, que es de **180 mg/dL**. Este signo lo podemos observar en un estudio llamado **examen general de orina** y nos ayuda a saber si hay algún problema en el metabolismo de los carbohidratos.

@Bioquimicaparanutris

Academia

www.academianutris.com

Blood Cells

- Presence of blood or RBC in urine Hematuria.
- Pathological condition.
- Causes –
- Glomerular nephritis, bleeding in UT , inflammation in urinary bladder and urethra.
- High BP

Red Urine

- Not a pathological condition
- Found in long distance runners
- Due to damage of blood vessel in the soil.

Bile pigments

- Associated with liver disease
- Diseases with heavy destruction of RBC.

- At liver, spleen and bone marrow.
- Pigmented portion of Hb, biliverdin bilirubin

- At liver
- Mix with Bile – bile pigments- helps in fat digestion.

- Bilirubin in urine- Bilirubinuria.
- Urobilinogen- Urobilinogen Uria.
- Urobilin- Urobilin Uria
- Causes –
- Pernicious anemia, hepatitis. etc

Casts

- Tiny masses of hardened materials
- Shaped like lumen of tubules.
- Eg leucocyte casts, erythrocyte casts, epithelial casts etc.
- Sure sign of glomerular nephritis.

Amino acid and ketone Bodies

- Blood amino acid level exceeds reabsorption capacity of tubules.
- E.g phenylketonuria (presence of phenylalanine) alkaptonuria (presence of alkapton)
- Detect the tubular reabsorption
- E.g Fanconi syndrome (due to Pb poisioning , inherited)
- Cystinuria (autosomal recessive disorder)

Nitrites

- Urinary tract infection bacteria that can reduce the nitrate to nitrite.
- Nitrate Reductase

- Thank you